YouTube Focus — Architecture Guide

Audience: another bot/engineer who must understand, modify, rebuild and re-publish this extension. Store id: `yt-focus` (type `extension`, target `panel-9`, current 1.4.0). Project folder: `/workspace/yt-focus-extension`. Runs in Panel 9 (`/workspace/mobile-browser`, Electron extension platform, see its ARCHITECTURE.md §10) and also loads unchanged as an unpacked MV3 extension in Chrome.

1. What it does

Popup with three switches (stored in `chrome.storage.local`, applied **live** to every open YouTube tab, desktop `www.youtube.com` and mobile `m.youtube.com`), plus one always-on behaviour:

Feature	Default	Behaviour
Switch account	always available	In comments-only mode a signed-in user keeps YouTube's own account menu: the avatar stays at the top-right and the bar has a **Switch account** button that opens the menu and jumps to its account/channel list (brand channels and *Add account* included). Works on desktop and mobile; in every other mode YouTube's UI is untouched. Never exposes Sign out (see next row).
Hide Sign out (`hideSignOut`)	ON by default (1.2.0: was always on)	The "Sign out" entry (any language in the table in `content.js`, logout hrefs, sign-out icons) is removed from the account menu / mobile sheets, including lazily rendered menus and SPA re-mounts; clicks/keys on a removed entry are swallowed.
`stopVideos`	off	Nothing plays or buffers: no autoplay on watch pages, no hover previews, no Shorts playback. `googlevideo.com/videoplayback` requests are refused before they reach the network.
`hideSuggested`	off	Hides the watch-page sidebar/related videos, end screens, the home feed and Shorts shelves/entry points (desktop and mobile). Search results, channel pages and the player stay.
`commentsOnly`	off	Watch pages show **only the comments**, full screen and scrollable (everything else incl. the player, masthead, metadata, sidebar is hidden; the video is also stopped). A thin bar offers "Switch account" (when signed in) and "Show video". Comment continuation loading, replies and sorting keep working.

2. Files

```
manifest.json     MV3. permissions: storage, declarativeNetRequest (1.4.0). background.service_worker =
background.js; declarative_net_request rule_resources = rules-lean.json (disabled by default).
background.js     (1.4.0) service worker: enables the 'lean' ruleset while commentsOnly is true. Not
needed/not run in Panel 9.
rules-lean.json   (1.4.0) 25 block rules (player, googlevideo, ads, telemetry, live chat), all
initiatorDomains youtube.com, never main_frame. host_permissions/matches: https://www.youtube.com/*,
https://m.youtube.com/*. action.default_popup = popup.html, options_page = popup.html
main-world.js     content script with "world":"MAIN" (page JS context, document_start): network/media
blocking
content.js        isolated-world content script (document_start): storage -> <html data-ytf-*>, Sign-out
sweeper, comments-only bar, mobile comment-sheet opener
styles.css        injected with content.js; every rule is gated by an <html data-ytf-*> attribute (hide /
focus), so toggling needs no re-injection
popup.html/.css/.js  the three switches; writes chrome.storage.local, mirrors changes made elsewhere
icons/            16/32/48/128 px;  store-icon-512.png = store listing icon
```

3. How the pieces talk

`chrome.storage.local {stopVideos, hideSuggested, commentsOnly, hideSignOut}` → `content.js` reads it at start and on `chrome.storage.onChanged` and mirrors the state to attributes on `<html>`: `data-ytf-stop`, `-hide`, `-comments`, `-stop-eff` (= stop OR comments), `-mobile`, `-watch` (current URL is a watch page, recomputed on `yt-navigate-finish` / `popstate` /mutations), `-focus` (watch page AND commentsOnly). `styles.css` and `main-world.js` only look at those attributes. After the first state is known `content.js` fires a `ytf-state` DOM event; `main-world.js` holds back video requests until then (1.5 s safety valve) so the first segment of an autoplaying page cannot slip through. In Panel 9 the popup lives in a small window; Panel 9 mirrors `chrome.storage.local` between all sessions (profiles, containers, private windows), so a toggle flips tabs everywhere.

3a. Switch account in comments-only mode (1.1.0)

Comments-only used to hide the whole masthead, which also removed the only way to reach YouTube's account menu. Now `content.js` `markAvatar()` finds the signed-in avatar button (desktop `ytd-masthead #avatar-btn` / `button[aria-label="Account menu"]` / a `ytd-topbar-menu-button-renderer` button holding an avatar `<img>`; mobile `ytm-mobile-topbar-renderer` `ytm-profile-icon` / `.topbar-menu-button-avatar-button`, never `.mobile-topbar-header-sign-in-button`), marks it `data-ytf-avatar`, marks every ancestor up to the masthead/topbar `data-ytf-chain`, and sets `<html data-ytf-acct>`. `styles.css` then shrinks `#masthead-container` / `ytd-masthead` / `ytm-mobile-topbar-renderer` to a 44 px fixed box at the top-right and hides every sibling that is not in the chain, i.e. only the avatar remains (no logo, search, Create, bell). Without an avatar (signed out) the masthead stays hidden as before and no Switch button is shown. YouTube's menu itself is untouched: it opens from the real avatar into `ytd-popup-container > tp-yt-iron-dropdown` (z-index 2202, above the bar/masthead at 2100) or the mobile bottom sheet (raised to z-index 2147483000 over the full-screen comment panel). The bar's **Switch account** button (`openSwitchAccount()`) clicks the avatar, waits for the menu item whose label is in the multilingual `SWITCH` set and clicks it; if the item cannot be found the menu simply stays open. If the account list is already open it does nothing.
Sign out stays removed in the Switch-account list: the sweeper runs synchronously from the MutationObserver (before paint) whenever something changes inside a menu/sheet root, plus CSS pre-hides `a[href*=Logout]` /sign-out-icon items, so the literal Sign out entry in the main menu and in the list never shows (also text-only variants without href). *Add account* (`accounts.google.com/AddSession`) and every channel/account row are kept.

4. Stop videos (`main-world.js`)

Patches, in the page world: `fetch` and `XMLHttpRequest` (reject/abort any `googlevideo.com/…/videoplayback` while `data-ytf-stop-eff` is set; counts blocked requests in `data-ytf-blocked`), `HTMLMediaElement.src` setter, `HTMLMediaElement.play()` (rejects with `NotAllowedError` like a blocked autoplay and pauses), and asks the YouTube player objects (`movie_player`, preview/Shorts players) to `stopVideo()`; `content.js` additionally pauses any `<video>` on `play` / `playing` and via a sweep (new videos appear constantly: previews, Shorts, SPA navigations). Because MSE buffers are fed by those fetches, blocking them means **no media bytes are downloaded** (verified with `webRequest` counting, see §7). Turning "stop" off on a page where requests were blocked reloads that page once so the player can start cleanly. The player is never removed from the DOM, so turning the switch off restores everything.

5. Hide suggested / comments only (`styles.css`)

Pure CSS keyed on the `<html>` attributes, scoped by page type so search results, channels and chips are not hit (`ytd-browse[page-subtype="home"]`, `ytd-watch-flexy`, mobile `[tab-identifier="FEwhat_to_watch"]`, `ytm-item-section-renderer[section-identifier="related-items"]` …). Comments-only hides the player, `#secondary`, metadata, panels, miniplayer etc. and stretches `ytd-comments#comments` to the page; the lazy wrappers around `ytd-comments` (`#below > .box`) are kept with `:has(ytd-comments)`. YouTube loads comments when the comment section scrolls near the viewport, so it must stay in layout (it is never `display:none`'d). On mobile the comment section is an engagement-panel bottom sheet: `content.js` (`modeSweep`) taps the "Comments" entry once the page shows it (and again if the user closes the sheet) and the CSS makes the sheet full screen.

6. Sign-out removal (`content.js`)

`sweepSignOut()` runs from a debounced `MutationObserver` on the whole document + a 1.5 s timer (menus are created lazily by Polymer/Lit and re-created on SPA navigation). It (1) walks known menu item tags (`ytd-compact-link-renderer` , `yt-list-item-view-model` , `ytm-menu-service-item-renderer` , ...) and hides those whose link points to a logout URL, whose icon is a sign-out icon, or whose label is a known "Sign out" string; (2) any `a[href*=logout]` inside menu-like containers; (3) leaf text nodes equal to a sign-out label inside popup containers/bottom sheets. Hidden = `display:none !important; aria-hidden; inert; data-ytf-signout` . A capture-phase listener swallows click/mousedown/pointerdown/touchstart/keydown on anything marked. Account **switching** and Settings stay available.

7. Tests (in the Panel 9 repo, network required)

`/workspace/mobile-browser/test/yt-focus-live.js` (desktop, real `https://www.youtube.com/watch?v=dQw4w9WgXcQ` , home, Shorts), `yt-focus-mobile.js` (Panel 9 mobile emulation, `m.youtube.com`), `yt-focus-switch.js` (Switch account on a fixture with the signed-in masthead/menu/list markup, `test/fixtures/yt-signedin-fixture.js`), `yt-focus-signout.js` (local fixture with the YouTube menu markup; real Sign out only exists when signed in). They load this folder with Panel 9's "Load unpacked", drive the popup like a user and assert: baseline video really downloads; live toggle pauses and stops `videoplayback` requests; fresh load with stop on makes 0 `videoplayback` requests with nothing buffered; clicking play does nothing; no hover-preview / Shorts playback; sidebar/home hidden and restored; comments-only layout, lazy comment loading while scrolling, "View replies"; sign-out variants hidden and clicks swallowed. Screenshots → `/workspace/mobile-browser/shots/yt-focus-*.png` . Run with `npm run test:ytf` in the Panel 9 folder.

8. Maintenance notes

- YouTube changes its markup often. Selectors live in `styles.css` / `content.js` ; when something breaks, run the tests, inspect with a probe (`webContents.executeJavaScript`) and adjust. The Sign out matcher is text/href/icon based on purpose, so it survives re-skins.
- Keep the three storage keys stable (users' settings); add new keys with defaults.
- Package: zip with `manifest.json` at the zip root (no folder wrapper), exclude `ARCHITECTURE.md` / `README.md` if you like (harmless to keep). `manifest.version` MUST equal the store version or Panel 9 refuses the install.
- Publish: `bash -lc '/workspace/appstore/bin/publish.sh yt-focus <version> yt-focus-<version>.zip "notes" --type extension --target panel-9 --platform any --source /workspace/yt-focus-extension --arch /workspace/yt-focus-extension/ARCHITECTURE.md'` (first release also needs `--name` , descriptions, `--icon` , `--screenshot` ; see the store STORE-GUIDE.md).
- Not verified: Switch account / Sign out removal on a real signed-in account (the markup was reproduced from YouTube's public element names/ids, not captured from a signed-in session; the signed-out real youtube.com was checked for the no-avatar path), Chrome desktop (only Panel 9/Electron 45), YouTube regions/experiments with different markup.

1.2.0 — 'Hide Sign out' is a switch (default ON)

- `chrome.storage.local.hideSignOut` (default `true`) is a fourth switch in the popup. `content.js` sets `data-ytf-show-signout` on `<html>` when it is OFF; `styles.css` keys both the `[data-ytf-signout]` rule and the pre-hide `:has(...)` rules on `html:not([data-ytf-show-signout])` , so the default (and the moment before storage has loaded) still hides it without a flash.
- `sweepSignOut()` returns immediately when the switch is off; `unhideSignOut()` removes `data-ytf-signout` , `aria-hidden` , `inert` and the inline `display` it had set, so turning the switch off shows Sign out again at once (storage listener + 1.5 s poll).
- Sign out is only ever hidden by this extension, only while installed and enabled. Panel 9 core contains no Sign out hiding (checked in `test/addaccount-checks.js` : Sign out is visible on the YouTube fixture without the extension). Per-account Sign out / Remove live in Panel 9's account switcher (`panel9://accounts`).
- Test: `test/addaccount-checks.js` (Panel 9 repo) runs `content.js` + `styles.css` against the signed-in YouTube fixture with a stubbed `chrome.storage` : hidden by default, visible after the switch is turned off, hidden again after it is turned on.

1.3.0 — hardening for Bean ("Comments only" must be solid)

Trigger: on a real run (Panel 9 2.3.10, 2 containers × 2 channels) Bean un-liked a comment it had just liked and used a channel that was not selected. The Bean side is fixed separately; this release makes Focus's half predictable. **No attribute, selector, event or DOM structure that Bean reads was changed** (contract below).

The contract Bean (and any other extension) can rely on — unchanged

What	Value
`<html data-ytf-comments>`	the "Comments only" **switch** is ON. Same on every youtube.com page (not page-scoped), so a tab on the home page still reports the real setting. Empty-string value.
`<html data-ytf-focus>`	the switch is ON **and** this is a watch page (`/watch`, `/live/<id>`, `/embed/<id>` in the top frame): the page is in comments-only layout. Removed the moment the URL leaves a watch page.
other `data-ytf-*`	`-watch`, `-stop`, `-hide`, `-stop-eff`, `-mobile`, `-show-signout`, `-acct`, `-blocked`, `-signout`, `-chain`, `-avatar`: same meaning as before.
`document` event `ytf-state`	fired when the state was (re)applied and now also whenever `-watch`/`-focus` flip.
`#ytf-bar`, `.ytf-btns`, `.ytf-show`, `.ytf-switch`	same ids/classes, still at most one `#ytf-bar`.
storage keys	`stopVideos`, `hideSuggested`, `commentsOnly`, `hideSignOut` (unchanged, only `chrome.storage.local`).
Only deliberate differences: `/watch` matching is exact (`/watchfoo` no longer counts); in a **subframe** (an embedded `youtube.com/embed/…` player on someone else's page) there is no watch layout, no `-watch`/`-focus` and no bar any more (stop-videos still applies there).	

One idempotent `sync()` (content.js)

Every `<html data-ytf-*>` attribute is derived by one cheap, synchronous function from *(stored switches, current URL)* and only writes differences. It is called from everything that can possibly signal a change: the Navigation API (`currententrychange` , fired synchronously *inside* `history.pushState/replaceState/back` , i.e. before YouTube paints the new page), `yt-navigate-start/finish/error` , `yt-page-data-updated` , `yt-page-type-changed` , `popstate` , `hashchange` , `pageshow` , `chrome.storage.onChanged` , **every DOM mutation** (so even if all navigation signals were missed, the next render fixes the attributes), and timers. Nothing depends on a timer (throttled in background tabs) or on focus/visibility events (they are only redundant extra triggers). A second `MutationObserver` with an attribute filter on Focus's own `<html>` attributes re-asserts them if anything removes or rewrites them.

- **document_start**: `-watch` / `-mobile` are set synchronously when the script runs (they need no storage); the switch-derived attributes right after the first `chrome.storage.local.get` resolves (a few ms, before the parser reaches `<body>` , measured by the tests: set at ~40–55 ms while a parser-blocking head holds `<body>` back for 300+ ms and first paint comes after). A stale `-focus` on a non-watch page is cleared even before storage answers.
- **SPA**: `main-world.js` wraps `history.pushState/replaceState` **only** in browsers without the Navigation API, and only to dispatch a plain `ytf-nav` DOM event after the original call returned (arguments/return value/exceptions untouched).
- `/signin?…pageid=…&next=` (Bean's channel switch) is just a non-watch youtube.com page: no `-focus` , no bar; the full `/watch` load that follows gets everything at document_start.
- **Storage consistency** (Panel 9 mirrors `chrome.storage.local` into every container/session and, while seeding a session, removes and re-sets keys): answers are sequenced (an older `get()` can never overwrite a newer one); values are type-checked (`'false'` is not truthy); a failing `get()` is retried with back-off; poll is 500 ms for the first 15 s of a page, then 1.5 s; **an ON→OFF change of "Comments only" is only believed if still OFF 250 ms later** (a transient "defaults" read cannot flip a comments-only page to the normal layout or reload it). OFF→ON and the other switches apply at once.
- **"Leaving stop mode reloads the page"** (needed because the starved player cannot restart) is now conservative: only on a *visible* watch page whose player was really blocked, only if the switch is still off after a second read, never for a flap, and a hidden tab waits until it is shown.

- Orphan guard: when the extension is reloaded/updated/disabled the old content script in open tabs detects `chrome.runtime.id` is gone and stops (all listeners via one `AbortController`, all timers, both observers). It leaves the attributes alone because the new copy may already own them. Previously the orphan kept running with a stale config next to the new copy.

The comment section is off limits

- Focus never clones, moves, re-inserts, hides (by JS) or clicks anything in or around the comment section, never touches a like button, `aria-pressed`, a click handler or hydration. Hiding is **CSS only** (`display:none` on siblings/wrappers, nothing is re-parented).
- The structural `MutationObserver` (menus appear lazily; masthead/avatar re-renders) **ignores every mutation whose target is inside the comments** (`ytd-comments`, `ytd-comment-thread-renderer`, `…-view-model`, `…-renderer`, replies, engagement bar, mobile comment/engagement panel), only ever calls `sync()` and its own sweeps, and only writes its own markers (`data-ytf-*`, `#ytf-bar`).
- Sign-out sweep guard (`safeToHide`): never marks anything inside the comments, never an element that *contains* comments / a like button / `video` / `ytd-player` / `ytd-watch-flexy`, never the menu/popup containers themselves. (A comment that merely links to `…/logout` is not an entry.)
- m.youtube.com's "open comments" helper only clicks the real entry point (`ytm-comments-entry-point-*`); the generic `aria-label*="comment"` fallback now skips anything inside a comment and anything labelled like/dislike/reply/pin/heart/report/edit/delete/share (it could previously match "Like this comment").
- Exactly one `#ytf-bar` (stray duplicates are removed); the bar is removed synchronously when leaving a watch page.
- CSS: every rule that can match an *ancestor* of `ytd-comments` (`#secondary`, `#panels`, engagement panels, `#columns > *`) now carries `:not(:has(ytd-comments))`, so a YouTube layout that puts the comments in a side column/panel cannot be hidden by us; `html[data-ytf-focus]{scroll-padding-top:56px}` so `scrollIntoView()` does not park a comment under the fixed 44 px bar. No rule uses `visibility/opacity/transform/content-visibility` on the comments.

Tests (`tests/`, not part of the zip)

- `tests/jsdom/*.test.js` — `cd tests && npm i && npm run test:jsdom` (23 tests, ~18 s): the real `content.js` / `main-world.js` in jsdom with a stubbed `chrome.*`; document_start, every navigation signal, missed signals, hidden tab with frozen timers, flapping/failing/out-of-order storage, comment-section mutation guard, duplicate detection, orphan, reload logic.
- `tests/e2e/` — `bash tests/run-e2e.sh`: the real Panel 9 (Electron 45; `P9_DIR`, default a pristine `git archive` of 2.3.10 at `/tmp/p9-ytf2310`) loads this folder with *Load unpacked*, drives the toolbar popup like a user, and runs against a local HTTPS mock of youtube.com (`mock-yt.js`, host mapped with `--host-resolver-rules`): fresh load with a parser-blocking head (attribute time vs `<body>` vs first paint), reload, YouTube-style SPA navigation incl. back/forward, `/signin?…pageid=…&next=` (302 and JS-redirect variants), a single like + a hydration-like re-render (one copy, one like, no un-like), hidden tab (really `visibilityState:'hidden'`, no throttling switches), two containers × 2 channels + a container created after the switch was set, Sign-out regression. `YTF_DIR=<old folder>` runs the same suite against another build.
- Not covered (needs real YouTube + a real signed-in account): see "Not verified" in the release notes.

Known Panel 9 2.3.10 limitation (found by the e2e, not a Focus matter)

A setting written within ~1 s before a brand-new container's session is first seeded can be missed by that session's storage host (Panel 9 pushes a change only to hosts already marked `seeded`, then seeds the new host from the older canonical copy and records it as in sync). That container then keeps the old value (e.g. comments-only OFF) until the next change. Focus reads exactly what `chrome.storage.local` holds, so it shows the stale value correctly. The e2e waits for the shared copy to settle before creating containers and prints `[diag]` host storage if it ever fails.

1.4.0 — lean "Comments only" and the linked-comment-only view

Goal (Oluchi): in Comments only mode every comment except the one linked by `lc=` is hidden *instantly*, with no flash, and the video never loads or shows; the page should *load* weightless, not load everything and then hide it.

Layers (outermost first)

1. **Network, Chrome (`background.js` + `rules-lean.json`).** A static declarativeNetRequest ruleset `lean` (disabled in the manifest) is enabled by the service worker on `chrome.storage` `commentsOnly === true` (also on install/startup/SW

start; the enabled state persists in Chrome, so at browser start the first request already sees it). Rules (all `initiatorDomains:["youtube.com"]`, subresource types only, `main_frame` never): `*.googlevideo.com`, `*.c.youtube.com`, `/s/player/` (base.js, www-player.css, remote/offline/captions/miniplayer/endscreen/annotations modules), `/embed/`, `/s/search/audio/`, `/api/timedtext`, lottie, cast_sender, `i.ytimg.com/an/` and the plain poster `…/vi/<id>/*default.jpg` (no `?sqp=`, so real thumbnails are untouched), `manifest.webmanifest`, doubleclick/googlesyndication/googleadservices, `/pagead/`, `/api/stats/`, `/ptracking`, `/pcs/activeview`, `youtubei/v1/log_event`, `youtubei/v1/player/heartbeat`, live chat. **Not** blocked (kept on purpose, see the allow-list test): `youtubei/v1/next` (comments), `youtubei/v1/comment/*` (like/reply actions), `youtubei/v1/browse`, ytcfg/app JS+CSS, fonts and the UI icon SVGs (`fonts.gstatic.com/s/i/…` are the like/dislike icons), `api/jnn/v1/GenerateIT` + `google.com/js/th` (attestation, tiny, left alone), `/signin`, `/getAccountSwitcherEndpoint`, accounts.google.com.
2. **Network, Panel 9 (branch `feat/ytf-lean-session-filter`, not published).** Panel 9's Electron extension platform has no declarativeNetRequest and no extension service worker, so `background.js` never runs there and the static `<script src=…/base.js>` tags in YouTube's HTML cannot be stopped from the page. Design: `src/ytf-lean.js` is the same list as a main-process `session.webRequest.onBeforeRequest` filter, active only while the YouTube Focus extension is enabled and its `commentsOnly` is true (read from the live extension-storage mirror `extMgr.peek`, no disk read) and only for requests whose top-level page is a youtube.com watch URL; main-frame requests are never cancelled. Electron allows **one** `onBeforeRequest` listener per session (the last registration silently replaces the earlier one) and the ad blocker already uses it, so `src/reqchain.js` owns the single listener and the ad blocker (`adblock.js`) and the lean filter register named filters in it. Unit test `test/ytf-lean-checks.js` (block list, allow list, scoping, chain coexistence). Verification: unit test + the Panel 9 e2e (51/51) was run both against pristine 2.3.10 and against this branch (`P9_DIR=/workspace/p9-ytf-lean bash tests/run-e2e.sh`, extension 1.4.0), i.e. the shared chain boots and does not break the ad blocker/extension; **not** verified: that the session filter itself cancels real YouTube requests inside Electron (no e2e for that yet).
3. **Page world (`main-world.js`).** Same list inside the page for everything started from script: `fetch`, `XMLHttpRequest`, `sendBeacon`, `HTMLScriptElement.src` / `HTMLLinkElement.href` / `HTMLImageElement.src` / `HTMLIFrameElement.src` and `setAttribute` of those (a blocked `<script>` gets `type="text/ytf-blocked"` and an `error` event so YouTube's loader does not hang), `HTMLMediaElement.load()` is a no-op, plus the 1.0–1.3 video request/ `play()` blocking. The mode is read synchronously: `localStorage['ytf-cfg']` (written by content.js whenever storage answers) until `<html data-ytf-known>` exists, then the attribute `data-ytf-comments`. Works in Panel 9 and in every browser. Cannot stop parser-inserted static tags (that is layers 1/2).
4. **CSS safety net (`styles.css`).** Everything below is keyed on attributes set synchronously by content.js at document_start, so nothing can paint first:
   - `html[data-ytf-pend]` — a watch page whose switch is not known yet (a few ms): player area collapsed (hide-by-default). `html[data-ytf-focus]` (known/provisional) and `html[data-ytf-lcf]` (lc present) collapse the same list: `#player` (incl. YouTube's server-rendered skeleton `<div id="player" class="skeleton flexy">` in `<body>`, `#player-wrap/#player-placeholder/#player-api`, and YouTube's `#watch-page-skeleton` with its ghost title/owner/comment placeholders), `#player-container*`, `#player-theater-container`, `#full-bleed-container`, `#cinematics`, `ytd-player`, `#movie_player`, `video`, `ytd-watch-metadata`, `#secondary` / `#related`, live chat.
   - `html[data-ytf-lcf]` — comments: `ytd-comment-thread-renderer:not([data-ytf-keep])` and the comments header/simple box are `display:none`; replies of a top-level target are hidden; for a reply target every reply view except `[data-ytf-target]` is hidden and the path to it is laid out even where YouTube keeps it collapsed (`#teaser-replies[hidden]`, `#expander[hidden]`); continuation renderers are 1 px / `visibility:hidden` (still intersecting, so the first batch loads) until `data-ytf-lcfound`, then `display:none` (no more pages are requested); ghost cards hidden.

Linked-comment state machine (`content.js`)

`lcFrom(href)` — `new URL(href).searchParams.get('lc')` then the hash (`#lc=…`, `#t=5&lc=…`, `#/watch?lc=…`); valid = `[A-Za-z0-9_.-]{10,300}`; only on watch pages in the top frame. `lcSync()` (called from `sync()`, i.e. from every navigation signal and every mutation) derives `data-ytf-lc` (lc present) and `data-ytf-lcf` = `pending` (mode unknown), `on` (mode on), absent (no lc / mode off / the user pressed *Show all comments*). `lcFilter()` runs from a dedicated MutationObserver (microtask after the DOM change, before paint): a comment is the target when its own timestamp link's `lc` equals the URL's (fallback: YouTube's `linked` attribute while the link is not stamped yet); the target gets `data-ytf-target`, its thread(s) `data-ytf-keep` (+ `-reply`), for a reply the ancestors inside the replies renderer `data-ytf-path`; YouTube keeps *two* copies of a linked reply (collapsed list + `#teaser-replies`): exactly one is shown (`data-ytf-dup` hides the other). Re-rendered comment sections are

detected (canonical element disconnected) and revealed again. `yt-navigate-start` carries the next URL (`detail.endpoint.commandMetadata.webCommandMetadata.url`), so the new page's lc is known before `location` changes.

Decisions / known limits

- Linked **reply** ("reply only"): ONLY the reply is shown. `lcFilter()` marks the reply `data-ytf-target`, the top-level thread `data-ytf-keep` + `-reply`, and **every ancestor between the reply and that thread** (replies renderer, `#replies`, `#teaser-replies` / `#expander…`) `data-ytf-path` (nothing else). CSS then hides every child of the thread and of each path element that is not itself on the path (`[keep][reply] > :not([path]):not([target])`, `[keep][reply] [path] > :not([path]):not([target])`): that removes the parent comment (`#comment-container` → header, body, action bar), the other replies, the reply toggles and "show more"; the path containers are force-laid-out (`display:block`, `visibility:visible`, auto height) and de-indented, so the reply's like button renders, is the hit target and is clickable. Marking happens in one synchronous pass together with `keep`, and the thread is `display:none` until then, so the parent is never laid out at any frame (mock probe: every insertion + every animation frame; real-YouTube probe). If the linked reply is somehow not inside a replies renderer it is treated as a top-level target (nothing hidden that could contain it).
- The first `youtubei/v1/next` response (comments batch, ~27 KB) also contains the related videos; it cannot be filtered at the network level, they are only hidden. The 2.2 MB YouTube application bundle and its 320 KB CSS are the critical path and are needed to render comments: **time-to-clickable-like-button did not improve** (≈4.1 s before and after (3 runs each, ±0.3 s noise) in headless Chromium; see the table below), the gain is bytes/requests/CPU and zero flash, not latency.
- Mobile (`m.youtube.com`): selectors for `ytm-comment-*` are included on a best-effort basis; not verified against real mobile markup.
- The mock + real-YouTube tests run unauthenticated; the like click is verified on the mock (DNR + page layer let `perform_comment_action` through) and the real like button is verified to exist and be clickable (`elementFromPoint` hits it), but no real like was sent.

Measurements (real YouTube, headless Chromium 156, unauthenticated, https://www.youtube.com/watch?v=dQw4w9WgXcQ&lc=… (top-level) and `…lc=parent.reply`; median of 3; for the reply row the flash probe also counts any layout of the parent comment as a violation)

A = no extension (comments loaded by scrolling), B = 1.3.0 mode on, C = 1.4.0 mode on. "flash violations" = frames/insertions where a non-linked comment or any player element had layout (capped at 50).

variant	like button clickable (ms)	bytes	requests ok / total	googlevideo requests that reached the network	flash violations	threads visible / loaded
A no extension	4153	6182 KB	205 / 212	4	50+	20 / 20
B 1.3.0 mode on	4159	4381 KB	170 / 170	1	50+	20 / 20
C 1.4.0 mode on	4167	3268 KB	132 / 140	0	0	1 / 20
B 1.3.0, reply link	4052	4590 KB	174 / 174	1	50+	20 / 20
C 1.4.0, reply link (reply only)	4190	3300 KB	134 / 142	0	0	1 / 20

(raw: `tests/live/out/*.json`, summary `tests/live/out/summary.md`; screenshots `tests/live/shots/`.)

Tests (1.4.0, not packaged)

- `tests/jsdom/lc.test.js` (12) + the 24 older tests (36 total): `cd tests && npm i && npm run test:jsdom` (d5 now allows the single `ytf-cfg` localStorage key).
- `tests/live/mock-e2e.cjs` — real Chromium (Playwright's `chromium`, headless) with the REAL extension loaded, against `mock-server.cjs` (local HTTPS, every youtube/googlevideo/doubleclick host mapped to it with `--host-`

`resolver-rules`, request log = ground truth for "what reached the network"). A probe installed before any page script records every insertion and every animation frame at which a non-highlighted comment node or any player node has layout. Scenarios A–H (B: reply link — parent comment never has layout at any insertion/frame, the reply never moves, its like button is the hit target, clicking it sends `perform_comment_action`; the mock DOM is thread > #comment-container > #comment, thread > #replies > replies renderer like the real one) (cold/warm cache, late target, reply, `#hash`, `p9bt`, SPA lc→lc→none→lc, missing target + timeout + button, mode off, mode on without lc, mode switched off live → reload, Bean flow: like click, `/getAccountSwitcherEndpoint`, accounts.google.com, `/signin` 302, reload) in two variants: DNR + page layer, and page layer only (ruleset disabled). Negative control: the same suite against 1.3.0 fails (`--ext /tmp/ytf130`). `node tests/live/mock-e2e.cjs --both`.

- `tests/live/measure.cjs` / `run-measure.sh` / `summarize.py` — real YouTube before/after (CDP network bytes/requests, like-button clickability, flash probe, early screenshots).
- `tests/run-e2e.sh` — Panel 9 e2e (51 checks) updated for the deliberate changes (`mock-yt.js` records attributes that were already set when its tracer starts; the comment-guard checks allow the five `data-ytf-keep/target/reply/path/dup` marks).